AI BASED REAL-TIME DETECTION OF CLONED AND HUMAN VOICES

Project seminar report submitted in partial fulfillment of the requirement for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

Under the esteemed guidance of

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CERTIFICATE

This is to certify that the B.Tech Project seminar report entitled **"AI-Based Real-Time Detection of Cloned and Human Voices"** is a bonafide work done by **K Manasa 22R11A05W5,Y Sahithi 22R11A05Y8, Manikanta Vedhas C 22R11A05W9** in partial fulfillment of the requirement of the award for the degree of Bachelor of Technology in "**Computer Science and Engineering**" from Jawaharlal Nehru Technological University, Hyderabad during the year 2025-2026.

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DECLARATION BY THE CANDIDATE

We, **K Manasa**, **Y Sahithi & Manikanta Vedhas C** bearing Roll Nos. **22R11A05W5, 22R11A05Y8, 22R11A05W9**, hereby declare that the project seminar report entitled **"AI-Based Real-Time Detection of Cloned and Human Voices"** is done under the guidance of **Mr. M Prashanth**, **Asst. Professor**, Department of Computer Science and Engineering, Geethanjali College of Engineering and Technology, is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**.

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With warm regards,
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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
DFD	Data Flow Diagram
API	Application Programming Interface
CNN	Convolutional Neural Network
MFCC	Mel Frequency Cepstral Coefficients
CSV	Comma Separated Values
HTTP	HyperText Transfer Protocol
MLP	Multi-Layer Perceptron
KNN	K-Nearest Neighbours
RNN	Recurrent Neural Network

1. ABSTRACT

The rapid growth of AI-based voice cloning has introduced significant risks such as impersonation, fraud, and misuse of synthetic speech in communication systems. Distinguishing real human voices from AI-generated cloned audio has become a critical need to ensure security and trust in digital interactions. This project proposes an AI-Based Real-Time Detection System that analyzes uploaded audio files and classifies them as either cloned or human. The system performs audio preprocessing, extracts key speech characteristics using Mel-Frequency Cepstral Coefficients (MFCC), and applies a trained Convolutional Neural Network (CNN) model for accurate classification. A Flask-based web interface is developed to allow users to upload audio files and instantly view detection results. The system also includes an email notification module to deliver results to users. The outcome of this project is a fast, lightweight, and reliable solution that enhances communication security by identifying artificially generated voices in real time.

2.INTRODUCTION

2.1 Background of Domain

Artificial intelligence has advanced rapidly in speech synthesis, enabling the generation of highly realistic cloned voices that sound almost identical to real human speech. While this progress has improved user experiences in many applications, it has also introduced significant risks, including impersonation, fraud, and misuse of synthetic audio. These challenges have created an urgent need for techniques that can reliably distinguish between genuine and AI-generated voices.

2.2 Motivation

With the rise of deepfake audio and voice-cloning tools, communication systems are becoming increasingly vulnerable. Fraudulent calls, identity theft, and unauthorized access through voice-based authentication are growing threats. To address these issues, a robust system capable of detecting cloned voices in real time is essential to maintain trust, security, and authenticity in voice-driven interactions.

2.3 Scope

The project focuses on building a complete pipeline for cloned-voice detection. It includes audio collection, preprocessing, MFCC-based feature extraction, and CNN-based classification. A real-time prediction interface is implemented using Flask, enabling users to upload audio files and instantly receive a classification result indicating whether the voice is human or cloned.

2.4 Applications

This system can be used in call verification, identity authentication, fraud prevention, and overall digital security. It is valuable in any domain where voice authenticity is critical, helping organizations and users safeguard communication channels and reduce risks associated with synthetic audio.

3. PROBLEM STATEMENT

3.1 Problem Statement

The rapid advancement of AI-generated voice cloning technology has made it increasingly difficult to identify whether a voice is genuine or artificially produced. These cloned voices are often highly convincing and can easily mimic real human speech patterns, leading to serious risks such as impersonation, financial fraud, and misuse in sensitive communication systems. Existing voice authentication techniques mainly focus on identifying the speaker but do not verify whether the voice itself is real or synthetically generated. Many available models work only on pre-recorded datasets, lack real-time processing capability, or high computational power. Additionally, there is a lack of simple, user-friendly systems that allow users to upload an audio file and instantly verify its authenticity. Therefore, there is a clear need for an efficient and lightweight system that can analyze audio, extract essential features, and accurately differentiate between cloned and human voices in real time.

3.2 Objectives

- To develop an AI-based real-time system for detecting and differentiating between cloned and human voices.
- To extract essential audio features using MFCC (Mel-Frequency Cepstral Coefficients) for accurate voice representation.
- To apply machine learning classifiers for reliable identification of genuine and AI-generated voices.
- To design an easy and interactive interface that enables users to upload audio files for instant detection.
- To improve voice-based communication security by preventing fraud and identity misuse through cloned voices.
- To ensure the system operates efficiently with minimal computational resources and supports real-time classification.

3.3 Existing System – Current Solutions & Limitations

Existing solutions in the field of voice analysis mainly concentrate on speaker authentication, where the goal is to verify who is speaking, rather than determining whether the voice itself is real or AI-generated. These systems fail to detect artificially cloned voices because they assume all input audio is genuine. Speech synthesis and recognition technologies used in virtual assistants have significantly improved voice quality, but they have also unintentionally increased the risk of voice cloning misuse. Although some online deepfake-audio detection tools exist, most of them are experimental research prototypes, require technical expertise to use, and are not suitable for real-time detection by general users. Furthermore, these systems often lack accuracy with short audio clips and do not provide simple upload-based verification. Due to these limitations, there is a need for a practical, real-time, user-friendly system capable of reliably distinguishing between cloned and human voices.

3.4 Proposed System – Contributions & Expected Benefits

The proposed system introduces a real-time solution for detecting and differentiating between cloned (AI-generated) and genuine human voices using MFCC-based feature extraction and CNN classification. Unlike existing systems that focus only on speaker identity, this model specifically targets voice authenticity, allowing users to verify whether an audio sample is artificially generated. The system provides a simple interface where users can upload audio files and instantly receive detection results, making it practical and accessible for non-technical users.

This project contributes a lightweight, fast, and real-time detection pipeline that does not require high computational resources. It enhances communication security by helping users identify potential voice-based threats and prevent impersonation or fraud. The proposed system also offers a structured workflow involving preprocessing, feature extraction, classification, and result display, ensuring reliable and consistent performance. Overall, it delivers a secure, user-friendly solution to address the growing misuse of AI-generated voice cloning.

4. LITERATURE REVIEW

Author & Year	Summary of Paper	Relevance to Proposed Project	Identified Gaps
Sonawane & Inamdar (2017)	Proposed a method to detect synthetic or spoofed speech using MFCC features and an SVM classifier.	Demonstrates MFCC as a strong feature for separating human and machine-generated voices.	Tested only on offline recorded data; no real-time detection support.
Patil et al. (2020)	Developed a speech verification system using pitch, energy, and MFCC features with a Decision Tree model.	Shows that simple ML models can capture differences in altered or modified speech.	Failed to detect AI-cloned voices and performed poorly in noisy environments.
Chakravarty & Dua (2023)	Created an MFCC-based hybrid ML model for detecting cloned or manipulated audio.	Validates MFCC + ML approach for identifying unnatural voice patterns.	Did not support real-time or streaming audio detection.
Alegre et al. (2014)	Explored spoofing detection using spectral and frequency-domain features with GMM classifiers.	Highlights the usefulness of frequency features for fake voice detection.	Low accuracy for short or real-world audio samples.
Kinnunen et al. (2017)	Studied synthetic speech detection using MFCC and CQCC feature variations.	Confirms cepstral features as strong indicators for differentiating manipulated speech.	Required very clean audio; weak against noisy or mixed input.
Li & Ma (2016)	Compared human and machine-generated voices using cepstral and spectral features.	Supports feature-based analysis for identifying AI-generated audio.	No direct user-level tool or instant prediction mechanism.

5. PROJECT METHODOLOGY

5.1 Overall Approach – Agile Methodology

For this project, the Agile development approach is followed to ensure flexibility, iterative improvement, and continuous testing of the voice detection system. Since the project involves model training, dataset updates, UI integration, and real-time prediction, Agile is suitable as it allows development in small, manageable phases (sprints). Each component— dataset creation, audio preprocessing, MFCC extraction, CNN model training, Flask integration, and UI testing—is developed and refined step-by-step.

Agile enables frequent evaluation of the model's performance and allows adjustments such as tuning parameters, improving preprocessing, or enhancing UI responsiveness. Regular feedback cycles help in refining both the classification accuracy and the user experience. This iterative process ensures that the system evolves continuously toward higher reliability and real-time performance. Overall, Agile supports faster development, easier debugging, and seamless enhancement of the cloned vs. human voice detection system.

5.2 Architecture Diagram

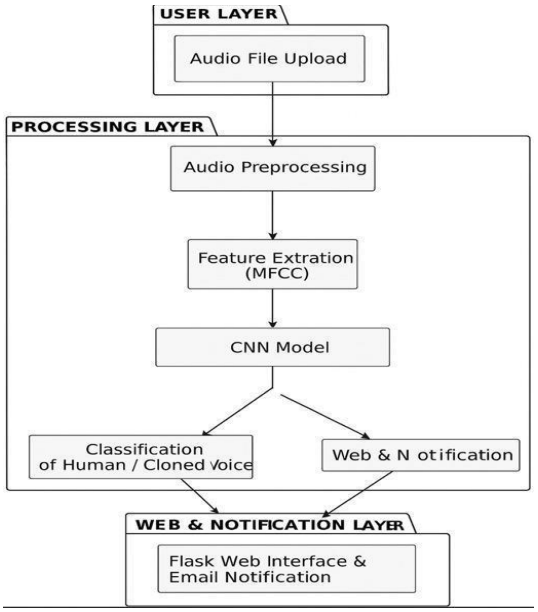

Fig- Architecture Diagram

5.3 Technology Stack

- Software:
  Web-based real-time voice detection system.
- Frontend:
  HTML, CSS, JavaScript for UI and audio uploading.
- Backend:
  Flask (Python) APIs for audio preprocessing, MFCC extraction, and prediction.
- Database:
  Local storage to save user data and prediction logs.
- Authentication:
  User login and access control (future scope).
- Audio Processing & AI:
  MFCC feature extraction and CNN classifier for detecting human vs. cloned voices.
- Deployment:

Standalone web application with real-time, low-latency prediction

6.SYSTEM DESIGN

6.1 Problem Analysis

The core problem is to identify whether an uploaded voice sample is genuine (human) or cloned (AI-generated) in real time. This requires converting a raw audio file, which can vary in format, length, and quality, into a consistent representation that a model can understand. The system must handle multiple sub-problems: accepting audio input from users, preprocessing it to remove noise and standardize duration, extracting meaningful features using MFCC, and then classifying the voice using a trained CNN-based machine learning model.

From a user perspective, the solution must be simple and interactive, allowing non- technical users to upload an audio file and quickly see whether the voice is cloned or human. Technically, the system should be lightweight enough to run on a normal laptop, provide predictions with low delay, and maintain reasonable accuracy to be useful in real- world scenarios such as basic fraud prevention and voice authenticity checking.

6.2 Design Diagrams

6.2.1 Use Case Diagram: This diagram represents the key interactions between the user and the AI-based voice detection system. It shows how the user uploads an audio file, while the system processes it by extracting MFCC features and classifying the voice as human or cloned. The result is then displayed back to the user.

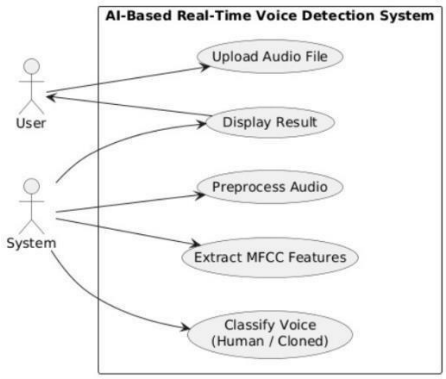

Fig-Use Case Diagram

6.2.2 DFD Diagram: This Data Flow Diagram shows the flow of information through the AI- based voice detection system. Audio is uploaded, preprocessed, and converted into MFCC features before being classified by the model. The final prediction is then sent back to the user as a displayed result.

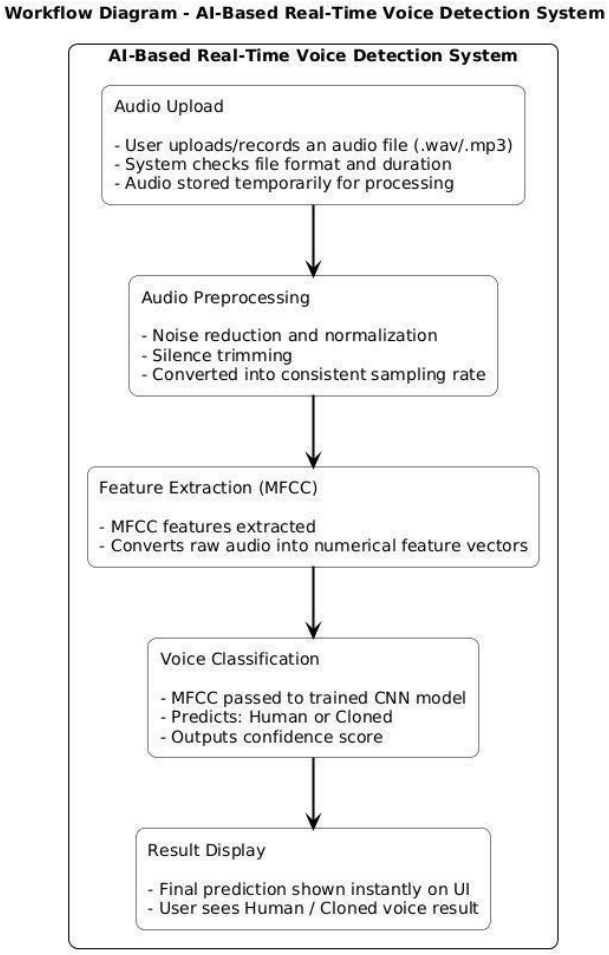

Fig- DFD Diagram

6.2.3 Class Diagram: This diagram represents the structure of the system by defining classes such as User, AudioFile, AudioProcessor, Classifier, and ResultDisplay. It shows the attributes and methods of each class and how they interact to process audio and generate the final prediction.

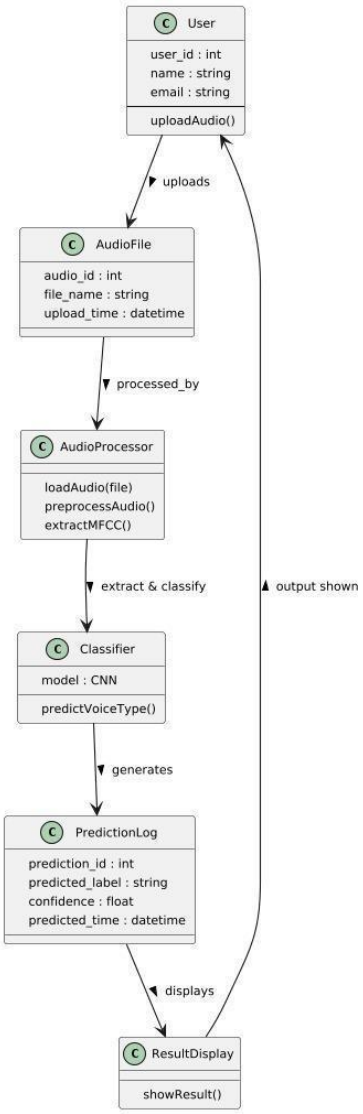

Fig- Class Diagram

6.2.4 Sequence Diagram: This diagram illustrates the flow of an uploaded audio file from the user's web interface to the Flask backend, where it is processed for noise removal and MFCC extraction. The processed features are passed to a CNN classifier, which predicts whether the voice is human or cloned, and the result is returned to the user.

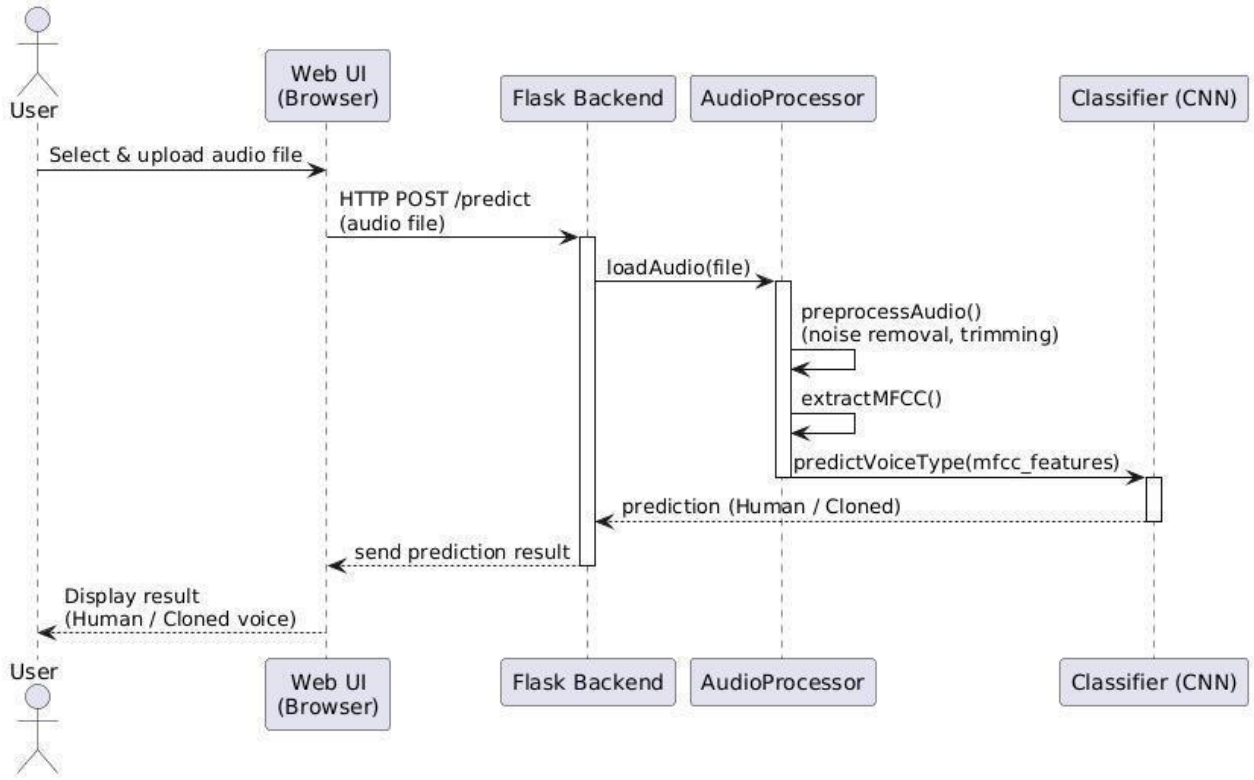

Fig- Sequence Diagram

6.3 Module Identification

1. Audio Input Module

This module allows users to upload or record an audio file. It validates file format, duration, and prepares the audio for processing.

2. Audio Preprocessing Module

Handles noise reduction, trimming of silence, normalization, and conversion of the audio into a uniform format suitable for feature extraction.

3. Feature Extraction Module

Uses MFCC (Mel-Frequency Cepstral Coefficients) to extract meaningful speech features that represent the audio's frequency characteristics.

4. Voice Classification Module

A trained CNN model processes the MFCC features and classifies the audio as Cloned Voice or Human Voice.

5. Result Display Module

Displays the detection result instantly to the user through a simple and interactive web interface built using Flask.

7. PROPOSED WORK PLAN

Phase	Timeline	Tasks	Deliverables
Phase 1: Requirement Analysis	Week 1	• Understand problem • Identify system needs • Define project objectives	Requirement specification document
Phase 2: Dataset Planning & Collection	Week 2	• Collect human voice samples • Collect cloned voice samples • Organize dataset into labeled folders	Complete and structured dataset
Phase 3: Audio Preprocessing	Week 3	• Noise removal • Trimming silence • Normalizing audio formats	Clean and standardized audio dataset
Phase 4: Feature Extraction (MFCC)	Week 4	• Extract MFCC features • Prepare feature matrices • Validate feature quality	MFCC feature dataset
Phase 5: Model Development (CNN)	Week 5	• Build CNN architecture • Train model • Tune hyperparameters	Trained CNN model with accuracy report
Phase 7: Frontend Interface Development	Week 7	• Design UI (HTML/CSS/JS) • Implement file upload • Display output results	User-friendly web interface

Phase 8: System Integration	Week 8	• Connect UI with backend • Integrate model predictions into UI • Handle input/output interactions	Fully integrated detection system
Phase 9: Testing & Evaluation	Week 9	• Test accuracy, speed & reliability • Fix errors • Validate predictions on new audio	Stable and optimized real-time detection system

8.CONCLUSION

During Semester-1, the focus was on understanding the problem of cloned voice detection and studying existing systems through a detailed literature review. Requirements were analyzed and the overall system architecture and methodology were planned. A custom audio dataset consisting of human and cloned voice samples was collected and organized. Initial preprocessing steps such as noise removal, trimming, and normalization were implemented to standardize the audio files. These foundational tasks helped establish the base structure needed for model development in the next phase.

In Semester-2, the remaining implementation work will be carried out. This includes extracting MFCC features from the preprocessed audio files, developing and training the CNN classification model, and building the complete Flask backend for real-time prediction. The frontend interface for audio upload and result display will be developed and integrated with the backend. The final steps will involve system testing, performance evaluation, optimization, documentation, and preparation of the final presentation and demo

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